CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CE/IT) Examination

November 2012

IT201 Database Management System

Date: 30.11.2012, Friday | **Time: 01:30 p.m. To 04:30 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Explain the advantages of database system over file processing system. [05]

(b) What is data independence? Explain the difference between physical and logical data independence. [02]

Q - 2 (a) Draw an ER Diagram for Media Library System. Media can be audio, video, books or magazines. [05]

(b) Explain the different mapping cardinalities of ER Model. [04]

OR

(b) Explain the different types of database languages in detail. [04]

(c) List the basic operators of relational algebra and explain any three in detail. [05]

OR

(c) Explain the concept of weak-entity set with suitable example. Differentiate the total participation and partial participation of entity set. [05]

Q - 3 (a) What is the need of Normalization process in database? Explain 1NF, 2NF, 3NF with suitable example. [08]

(b) Given FD's for relation R {A, B, C, D, E, F}, Find the closure of FD set by applying Armstrong's axioms. [03]

$F = \{A \rightarrow B, \quad A \rightarrow C, \quad CD \rightarrow E, \quad CD \rightarrow F, \quad B \rightarrow E\}$

OR

(b) Given FD's for relation R {A, B, C, D, E, F}, Find the closure of FD set by applying Armstrong's axioms. [03]

$F = \{A \rightarrow B, \quad AB \rightarrow C, \quad D \rightarrow AC, \quad D \rightarrow E\}$

(c) Find the canonical cover for relation R {A, B, C}: [03]

$F = \{A \rightarrow BC, \quad B \rightarrow C, \quad A \rightarrow B, \quad AB \rightarrow C\}$

OR

(c) For Relation R = {A, B, C}, where $F = \{A \rightarrow B, B \rightarrow C\}$. Check whether following decompositions are lossless join decomposition or not. [03]

1) R1 = (A, B), R2 = (B, C)
2) R1 = (A, B), R2 = (A, C)

SECTION – II

Q - 4 (a) Compare view serializability and conflict serializability with example. [05]

(b) Explain the types of lock and their compatibility matrix. [02]

Q - 5 (a) Explain the properties and states of transaction in brief. [06]

OR

(a) Explain the shadow copy method of transaction in detail. What are the advantages of concurrent executions in transaction [06]

(b) Explain deadlock detection techniques in detail. [04]

OR

(b) Differentiate between wait-die and wound-wait scheme of deadlock prevention techniques. [04]

(c) Define the following terms: [04]

i. Starvation

ii. Cascading Rollback

iii. Serializability

iv. Functional Dependency

Q - 6 (a) Consider the following database. [08]

Emp(eid, ename, salary, dept_id)

Dept(dept_id, dname)

1) Give a Relational Algebra Expression for following queries:

i. List the employee having salary > 10000 and working in 'IT' department.

ii. List the average salary of employee from each department having salary > 5000

2) Give a SQL Expression for following queries:

i. List the Employee information in descending order of ename with nulls first.

ii. List the all the information of employee and department of those employee having "_" as a substring in employee id.

(b) Explain three concurrency problems in database system with solution. [06]

OR

(b) What is Log based recovery? Explain any two Log based recovery techniques with example. [06]

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